BAIER–ZHAO DISPERSION SCALE BARRIERS FOR QUADRATIC-PRIME REPRESENTATIONS $n = ap^2 + b$

ARGUS OPEN RESEARCH PROJECT

Abstract. Erdős Problem #676 asks whether every sufficiently large integer n can be written in the form $n = ap^2+b$ with p prime, $a \geq 1$, and $0 \leq b < p$. We prove that on the signed covariance dispersion route at exponent $\kappa < 1/2$, the Baier–Zhao large-sieve bound for primes in quadratic progressions supplies only $Y^{34+o(1)} \log Y$ against the required Y^2 saving, excluding that dispersion interface without resolving the conjecture.

1. Main Scale Analysis

Proposition 1.1. *At remainder threshold $b < p^\kappa$ with $\kappa < 1/2$, the off-diagonal signed covariance sum over moduli $a \leq X/Y^2$ requires cancellation below X^2/Y, whereas the unconditional Baier–Zhao quadratic large-sieve estimate yields an upper bound of order $Y^{34+o(1)} \log Y$, leaving an unconditional exponent gap on that analytic route.*

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